

Data Mining Report

Phase 3

[1] Problem

Housing prices vary significantly due to differences in property attributes such as price, bedrooms, grade, renovated, and other structural and condition-related factors. These variations make manual price estimation unreliable and inconsistent, especially when dealing with large volumes of properties.

Because of this complexity, there is a need for a data-driven classification approach that groups houses into pricing categories based on their attributes rather than subjective market judgment. This problem is important because accurate classification helps real-estate analysts, buyers, and sellers make more informed decisions, understand market positioning, and reduce pricing bias.

This project aims to classify properties into four pricing categories (quartile_zone) by analyzing feature patterns and determining how each attribute contributes to property value. The resulting classification provides a more objective mechanism for evaluating property prices and understanding market segmentation.

[2] Data Mining Task

This project is formulated as both a classification and clustering task. The classification task will be performed using a supervised learning approach, where the goal is to assign each house to its corresponding price quartile based on its features. The target variable `quartile_zone` contains four predefined price categories, making classification an appropriate choice for predicting pricing levels rather than estimating exact values.

In addition, a clustering task will be applied using unsupervised learning techniques to group houses based on similarity in their attributes without using the target variable. This helps reveal natural patterns within the dataset and allows comparison between the formed clusters and the actual labeled quartiles. Using both approaches provides complementary perspectives: classification supports prediction, while clustering supports pattern discovery and exploration.

[3] Data

The selected dataset, obtained from Kaggle under the name House Price Prediction Treated Dataset, initially contains 6700 instances and 14 attributes encompassing both numerical and categorical variables. The features describe multiple aspects of residential properties, including listing date, price, number of bedrooms and bathrooms, living area size, house grade, basement availability, renovation status, view quality, overall condition, lavatory availability, and whether the house is a single-floor home. The target attribute, `quartile_zone`, categorizes each house into one of four price quartiles (1 to 4). The dataset demonstrates a balanced class distribution across these quartiles, making it appropriate for classification tasks and enabling the development of models that predict a home price category based on its features.

During Phase 2, the dataset underwent a comprehensive preprocessing stage aimed at enhancing data quality and ensuring readiness for machine learning analysis. This process involved cleaning and filtering the data to remove instances containing missing, inconsistent, or invalid values. After preprocessing, the dataset consisted of 6576 valid instances and a refined set of 11 attributes, all free from duplicates and structural inconsistencies.

Preprocessing steps included feature refinement, where non-essential or redundant attributes were removed to focus on variables most relevant to the prediction task. Several attributes describing structural and qualitative house characteristics—such as `grade`, `living_in_m2`, `has_basement`, `renovated`, `nice_view`, and `perfect_condition`—were retained based on their predictive value. Boolean attributes were standardized into consistent binary formats (True/False), while numerical features such as `price` and `living_in_m2` were normalized or scaled to ensure uniform ranges and prevent magnitude bias. The date field was also cleaned and unified into a consistent format.

The target variable `quartile_zone` remained unchanged and continued to represent the four price-based categories. Its balanced distribution persisted after cleaning, supporting stable and unbiased model training.

Overall, the preprocessing phase produced a structured, high-quality, and analysis-ready dataset. The resulting dataset from Phase 2 provides a reliable foundation for applying classification techniques and developing accurate predictive models in subsequent phases.

[4] Data preprocessing

The dataset underwent comprehensive preprocessing to ensure data quality, consistency, and readiness for modeling. The following steps were applied:

Handling Missing Values: The dataset was checked for missing values, and none were found. Therefore, no imputation was required.

Standardization of Numeric Features: Numeric attributes with different scales, including price, living_in_m2, bedrooms, and real_bathrooms, were standardized using StandardScaler. This transformed the features to have a mean of 0 and a standard deviation of 1, which prevents bias caused by large differences in feature scales.

Feature Selection: The ANOVA F-test was applied to identify features with the weakest relationship to the target variable `quartile_zone`. The features `month`, `bedrooms`, and `real_bathrooms` were found to be least important and removed. This step reduced noise and improved dataset relevance.

Outlier and Noise Removal: Extreme values in numeric features, particularly price and living_in_m2, were detected using the Interquartile Range (IQR) method and removed. This step ensured that unusual or extreme observations do not skew the analysis or model training.

Preservation of Boolean Features: Categorical boolean features, such as `has_basement`, `renovated`, `nice_view`, `perfect_condition`, `has_lavatory`, and `single_floor`, were retained as-is, since they contained no missing values and did not require scaling.

Resulting Dataset: After preprocessing, the dataset became cleaner, more consistent, and reliable. Numeric features were standardized, irrelevant, or weak features were removed, and extreme outliers were mitigated.

[5] Data Mining Technique

Classification:

For the classification task, we use a Decision Tree classifier from the scikit-learn library (`sklearn.tree.DecisionTreeClassifier`) to predict the house price category `quartile_zone`. This model is suitable because it can handle numerical features, is easy to interpret, and supports different impurity measures. We train six Decision Tree models using three train-test split ratios (70/30, 80/20, 90/10) and two attribute selection criteria: Gini index and Entropy (Information Gain). The models are implemented using `train_test_split` from `sklearn.model_selection` and evaluated with accuracy and the confusion matrix from `sklearn.metrics`, then visualized using `plot_tree` and `matplotlib`.

Clustering:

In this project we applied K Means clustering to identify groups of similar houses in the dataset. K Means was chosen because it is efficient and easy to implement and works well with numerical features such as price, size, grade, and number of bedrooms.

[6] Evaluation and Comparison

Classification:

confusion matrix:

Split: 70/30 | Criterion: gini

Accuracy: 0.5297

Confusion Matrix: [

[[338 126 21 5] [147 264 116 53] [27 109 237 137] [9 48 130 206]]

Split: 70/30 | Criterion: entropy

Accuracy: 0.5124

Confusion Matrix: [[326 137 21 6] [149 265 117 49] [22 123 234 131] [12 67 128 186]]

Split: 80/20 | Criterion: gini
Accuracy: 0.5243
Confusion Matrix: [[227 85 10 5] [96 182 73 36] [17 86 155 82] [8 27 101 126]]

Split: 80/20 | Criterion: entropy
Accuracy: 0.5243
Confusion Matrix: [[227 81 15 4] [86 176 90 35] [13 78 166 83] [4 36 101 121]]

Split: 90/10 | Criterion: gini
Accuracy: 0.541
Confusion Matrix: [[112 44 7 1] [49 87 44 13] [6 41 90 33] [2 17 45 67]]

Split: 90/10 | Criterion: entropy
Accuracy: 0.541
Confusion Matrix: [[114 47 3 0] [47 90 39 17] [8 35 90 37] [4 18 47 62]]

Accuracy Comparison:

Split	Criterion	Accuracy
70/30	gini	0.5297
70/30	entropy	0.5124
80/20	gini	0.5243
80/20	entropy	0.5243
90/10	gini	0.5410
90/10	entropy	0.5410

Best Algorithm per Partition

- 70/30 Split: Gini achieved the highest accuracy (0.5297).
- 80/20 Split: Both Gini and Entropy achieved the *same* accuracy (0.5243).
- 90/10 Split: Both Gini and Entropy achieved the highest overall accuracy (0.5410).

Overall Best Algorithm

Across all partitions, both Gini and Entropy performed equally well. The highest accuracy achieved was 0.5410, and both criteria reached this score in the 90/10 split.

Performance Interpretation

The accuracy values are around 0.50 because the dataset contains four classes, making the classification task more challenging. Both Gini and Entropy perform similarly, which is expected for this type of numerical data.

Confusion Matrix Insights

- The model sometimes confuses neighboring quartile zones.
- IT is normal because the house features can be similar across close price ranges.
- The model still shows a reasonable ability to distinguish between classes.

Clustering:

Clustering was used to uncover hidden groups in the dataset without relying on the quartile labels using Elbow method and silhouette method to daisied the best K ,after that we applied kMeans

K Means was applied using K = 2, 3, 4

Clustering Comparison:

K	Silhouette	WCSS
2	0.428900	9892.738000
3	0.322000	7341.567600
4	0.316300	6285.218300

Discussion – Clustering Results

1. Best K Value Based on the majority rule (Silhouette Score, WCSS, and visualization), the best number of clusters is $K = 2$.

2. Interpretation

- $K = 2$ gives the clearest separation between groups.
- Increasing K reduces WCSS but does not improve the Silhouette Score.
- This means adding more clusters does not necessarily improve cluster quality.

3. Insights

- K-Means successfully grouped the dataset into meaningful clusters.
- Houses in the same cluster share similar characteristics such as size, quality, and condition.
- These clusters help reveal hidden patterns that are not obvious from classification.

[7] Findings and Discussion

Classification Findings:

The classification task successfully predicted the housing price quartile (Q1–Q4) based on numerical features.

Among all tested classification models, the Decision Tree using the Gini criterion with an 80/20 split achieved the best performance.

The final Decision Tree visualization revealed clear rules and feature thresholds that influence the price quartile. The most impactful attributes were:

- OverallQual – higher overall quality increases the likelihood of falling into Q3 or Q4.
- GrLivArea – larger living area is associated with higher price quartiles.
- YearBuilt – newer homes tend to fall in higher quartiles.
- GarageCars – houses with larger garages belong to higher price segments.

These patterns provide interpretable and actionable insights, showing why the model assigns a house to a certain quartile.

Problem Solution:

The classification model provides a direct solution to the main problem:

Predicting the exact quartile zone of a house based on its features.

This makes classification the technique that solves the supervised prediction problem.

Clustering Findings:

Clustering was used to uncover hidden groups in the dataset without relying on the quartile labels.

K-Means was applied using $K = 2, 3, 4$ Using the majority rule (Silhouette Score, WCSS, and visual separation), $K = 2$ was identified as the optimal number of clusters.

The clustering results revealed two meaningful housing segments:

- Cluster 0: Houses with larger areas, higher quality scores, and better overall features.

- Cluster 1: Houses with moderate or lower values across the same attributes.

These patterns show that the market naturally divides into two main categories based on structural characteristics.

The cluster visualization confirmed:

- Clear separation between the two groups.
- Strong internal similarity within clusters.
- Structural features heavily influence grouping.

Problem Solution:

While clustering does not predict quartile labels, it provides a different type of solution:

Identifying natural market segments and revealing hidden structure within the dataset.

This supports a better understanding of real estate patterns and pricing behavior.

Comparison Between Classification and Clustering:

When comparing both data mining techniques, the classification approach clearly provides the best solution to the main problem, which is predicting the housing price quartile (Q1–Q4).

The Decision Tree using the Gini criterion (80/20 split) is the most effective model because it directly performs supervised prediction and achieves the highest accuracy.

On the other hand, clustering does not solve the prediction task; instead, it serves as an exploratory technique that reveals meaningful patterns and natural groupings in the housing dataset.

Therefore, the best-performing technique for solving the problem under study is Classification, while Clustering provides supportive insights but not a predictive solution.

Relation to the Research Paper:

The selected research paper highlights that structural housing attributes, such as overall quality, living area size, and construction characteristics, play a major role in determining house prices. It also emphasizes the importance of combining predictive models with unsupervised techniques to better understand market structure.

Our findings strongly align with the paper's conclusions:

- The most influential attributes identified by the Decision Tree (OverallQual, GrLivArea, YearBuilt) match the key factors emphasized in the research.
- The clustering results support the paper's observation that houses naturally group into meaningful segments based on quality and size.
- Both the classification and clustering results reinforce the idea that machine learning techniques can reveal clear patterns and provide valuable insights into housing price behavior.

Overall, the outcomes of this project are consistent with the research paper and validate the importance of structural features and pattern discovery in real estate analysis.

[8] References

[1] A. Aravind, "House Price Prediction Treated Dataset" [Online dataset]. Kaggle.

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[2] L. Breiman, J. Friedman, R. Olshen, and C. Stone, *Classification and Regression Trees*. Belmont, CA: Wadsworth, 1984.

[3] Y. Zhao, J. Zhao, and E. Y. Lam, "House Price Prediction: A Multi-Source Data Fusion Perspective," *Big Data Mining and Analytics*, vol. 7, no. 3, pp. 603–620, Sep. 2024, doi: 10.26599/BDMA.2024.9020019.